

Discrete-Time Plant Modeling for Efficient Validation of Control Systems

1 Introduction

Control systems for electrical drives and grid-connected converters are often designed using linearized models in the discrete domain with tools such as MATLAB `sisotool`. While this approach is effective for preliminary controller synthesis, the resulting closed-loop responses frequently deviate from detailed simulation results obtained from platforms such as PLECS or full MATLAB–Simulink models.

This discrepancy primarily arises due to:

- Neglect of cross-coupling terms in multi-variable systems,
- Assumptions of independent state variables,
- Simplified disturbance modeling during controller design,
- Differences between idealized plant models and actual switching or averaged dynamics.

Although comprehensive time-domain simulations provide higher fidelity, they are computationally intensive and slow to iterate, especially during PI controller tuning. This motivates the need for an intermediate validation framework that balances modeling accuracy with simulation efficiency.

2 Motivation for Discrete-Time Plant Modeling

Most modern digital control systems operate at sampling intervals in the order of $50\text{ }\mu\text{s}$ to $200\text{ }\mu\text{s}$. At such sampling rates, it is often reasonable to approximate the plant dynamics directly in the discrete domain, instead of relying on continuous-time simulation with numerical solvers.

The key idea explored in this work is:

Discretize the plant model itself and validate the controller using a fully discrete closed-loop representation.

This approach offers several advantages:

- Drastically reduced simulation time,
- Accurate assessment of feedforward and cross-coupling compensation,
- Direct evaluation of disturbance rejection performance,
- Seamless portability of the model to embedded controllers,
- Possibility of using the discrete plant as a real-time digital simulator (RTDS).

While inverter switching dynamics and non-idealities are not explicitly modeled, the framework serves as a powerful first step in controller validation.

3 PMSM Model in the dq -Reference Frame

The surface-mounted PMSM model in the synchronous rotating reference frame is given by:

$$\frac{di_d}{dt} = \frac{1}{L_d} (v_d + \omega_e L_q i_q - R i_d), \quad (1)$$

$$\frac{di_q}{dt} = \frac{1}{L_q} (v_q - \omega_e L_d i_d - \omega_e \lambda_m - R i_q), \quad (2)$$

$$\frac{d\omega_m}{dt} = \frac{1}{J} (T_e - T_L - B\omega_m). \quad (3)$$

In the Laplace domain, neglecting initial conditions, the current dynamics can be written as:

$$i_d(s) = \frac{1}{R + sL_d} (v_d + \omega_e L_q i_q), \quad (4)$$

$$i_q(s) = \frac{1}{R + sL_q} (v_q - \omega_e L_d i_d - \omega_e \lambda_m). \quad (5)$$

A key observation is that each state equation is fundamentally a first-order system with additional coupling and disturbance terms.

4 Generic First-Order System Representation

Consider a generic first-order plant:

$$Y(s) = \frac{1}{\alpha s + \beta} X(s), \quad (6)$$

where α and β are positive constants.

This structure is common across:

- Electrical current dynamics,
- Mechanical speed dynamics,
- Grid-side filter models.

Such a unified representation simplifies the discretization and implementation process.

5 Discrete-Time Approximation Using Tustin Method

Applying the Tustin (bilinear) transformation:

$$s \approx \frac{2}{T_s} \frac{z - 1}{z + 1}, \quad (7)$$

the discrete-time realization of the plant becomes:

$$y[n] = G_1 y[n - 1] + G_2 (x[n] + x[n - 1]), \quad (8)$$

where:

$$G_1 = \frac{\alpha - 0.5T_s\beta}{\alpha + 0.5T_s\beta}, \quad (9)$$

$$G_2 = \frac{0.5T_s}{\alpha + 0.5T_s\beta}. \quad (10)$$

These equations are purely algebraic and require only:

- Current input sample,
- Previous input sample,
- Previous output sample.

Hence, they are ideally suited for implementation using basic arithmetic blocks and persistent variables.

6 Implementation in MATLAB–Simulink

The discrete plant equations were implemented using standard Simulink blocks and MATLAB Function blocks with persistent states. Each dynamic element of the PMSM model was replaced by its discrete-time counterpart, while explicitly retaining:

- Cross-coupling terms,
- Feedforward compensation,
- Load torque disturbances.

The complete closed-loop PMSM control system, including current and speed controllers, was simulated entirely in the discrete domain.

This approach enabled rapid tuning of PI gains and clear isolation of control-loop interactions without the computational overhead of continuous-time solvers.

7 Validation Against Full-Order PLECS Simulation

To verify the fidelity of the proposed discrete-time plant modeling approach, a direct comparison was performed against a high-fidelity PLECS simulation.

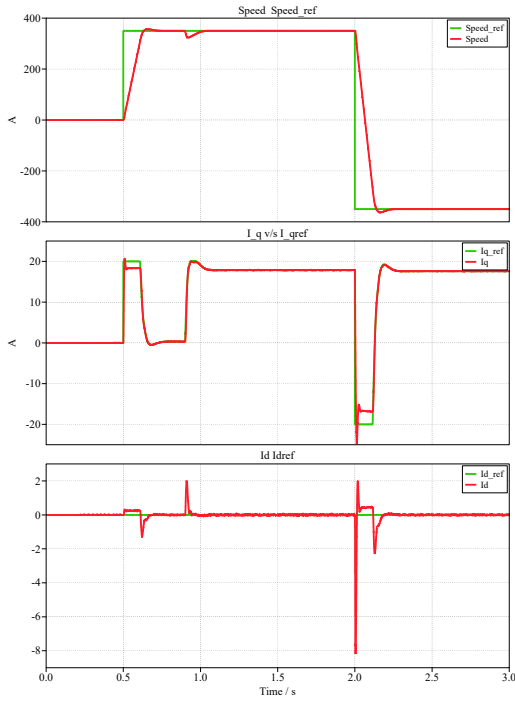
7.1 Simulation Setup

Two simulation environments were considered:

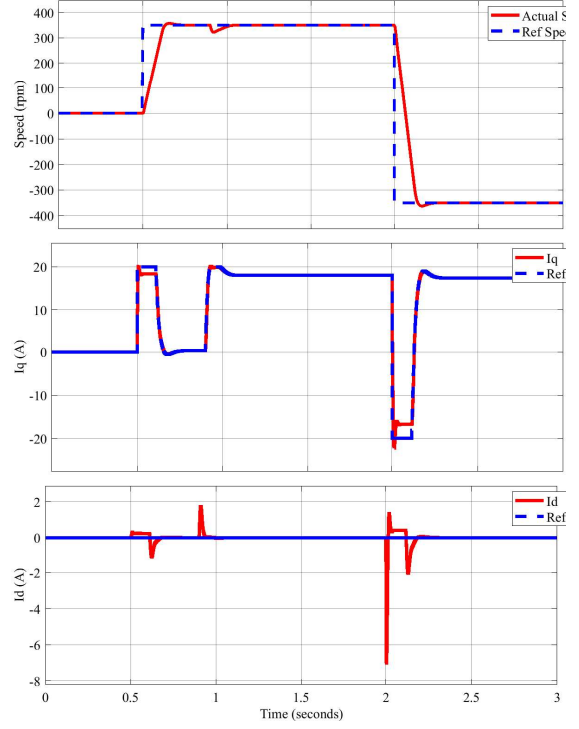
- **P1:** A detailed PLECS model incorporating the full PMSM dynamics, inverter effects, and continuous-time plant behavior.
- **P3:** A MATLAB–Simulink model based entirely on the discretized plant equations derived in this work, executed at the controller sampling time.

Both models were operated under **identical conditions**

7.2 Comparative Results



(a) P1: Full-order PLECS simulation including continuous-time PMSM dynamics.



(b) P3: MATLAB discrete-time plant model executed at the controller sampling interval.

Figure 1: Comparison of closed-loop PMSM responses under identical operating conditions. Both models employ identical controller parameters, sampling time, and reference profiles. The close agreement in speed, i_q , and i_d responses demonstrates that the proposed discrete-time plant model accurately captures the dominant system dynamics for control validation purposes.

Minor discrepancies, primarily during fast transients, can be attributed to inverter switching dynamics and continuous-time effects that are intentionally neglected in the discrete model.

7.3 Discussion

This result supports the following important conclusions:

- The discrete-time plant is adequate for validating controller structure and tuning.
- Cross-coupling and feedforward effects are correctly preserved in the discrete formulation.
- Disturbance rejection behavior observed in the discrete model is representative of the actual system.

Consequently, the proposed framework provides a computationally efficient alternative to full-order simulations during early-stage controller development, without sacrificing essential dynamic behavior.

8 Conclusion

This report presented a systematic approach for validating discrete-time controllers using discretized plant models. By exploiting the inherent first-order nature of many electrical and mechanical subsystems, accurate and computationally efficient closed-loop simulations can be achieved.

The methodology significantly reduces development time while preserving essential system dynamics, making it a valuable tool for control engineers working on motor drives and grid-connected systems.